

REVIEW

Why Are Some Individuals More Resilient to Chronic Stress Than Others? A Comparative Review of the Biological Mechanisms Underlying Stress

Resilience and Stress Susceptibility

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Abstract

Chronic stress has been reported to induce long-lasting changes in cognitive, emotional and behavioral functions. However, individuals subjected to similar stressors can present vastly different outcomes. This review aimed to identify the key cellular, molecular and neural mechanisms by which stress-resilient and stress-susceptible individuals are characterized and how these differences could affect their long-term cognitive functions. For this review, we conducted a narrative and comparative literature review to identify relevant studies. We searched PubMed and Google Scholar using various combinations of keywords and identified studies published after 2010. We selected studies that reported quantitative data and evaluated the methodological quality of the studies. We organized the included studies in a Literature Review Matrix and analyzed the evidence along several dimensions including protein regulation, the HPA-axis and stress signaling, epigenetic regulation, blood-brain barrier and neurovascular function, neuroinflammation and neural plasticity. Overall, studies identified as susceptible phenotypes were characterized by stress signaling, increased inflammatory signaling, altered protein and epigenetic regulation of genes and impaired blood-brain barrier function with corresponding changes in neural plasticity that may lead to cognitive and behavioral deficits. In contrast, studies that identified resilient individuals revealed adaptive responses to chronic stress including unique molecular signatures, containing inflammatory signaling, intact neurovascular function and neural plasticity. Importantly, several of the studied mechanisms appeared to interact with each other, e.g. by affecting inflammatory signaling, blood-brain barrier permeability and the neural environment. In summary, resilience seems to be brought about by coordinated regulation of different biological systems that together limit the impact of chronic stress and protect cognitive and behavioral functions.